LAB EXERCISE 03 GPIO LED CONTROL

Light the BLUE and RED LED alternately

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Project	**GPIO_02**
Student Number	**2442130210**
Target MCU	**STM32L431RCTX**
Main Source File	**07_NosPrg / main.c**
Exercise	**BLUE ↔ RED alternating LED control**

Prepared as a clean, teacher-ready laboratory submission. The report documents the objective, control logic, source-code modification, test procedure, expected result, and verification checklist.

1. Objective

Based on the supplied GPIO project, implement a simple alternating LED sequence. The BLUE LED and RED LED must never remain ON together. The program should switch between the two colors repeatedly with a visible delay.

Step 1	Step 2	Step 3	Step 4	Repeat
BLUE ON	Delay	RED ON	Delay	

Expected sequence

BLUE → RED → BLUE → RED → …

2. Project Environment

- IDE: AHL-GEC-IDE V4.53
- Project archive: GPIO_02
- Target MCU: STM32L431RCTX
- Application source: 07_NosPrg/main.c
- GPIO driver: 03_MCU/MCU_drivers/gpio.c and gpio.h
- LED control uses the project macros LIGHT_RED and LIGHT_BLUE.

3. Project Structure Evidence

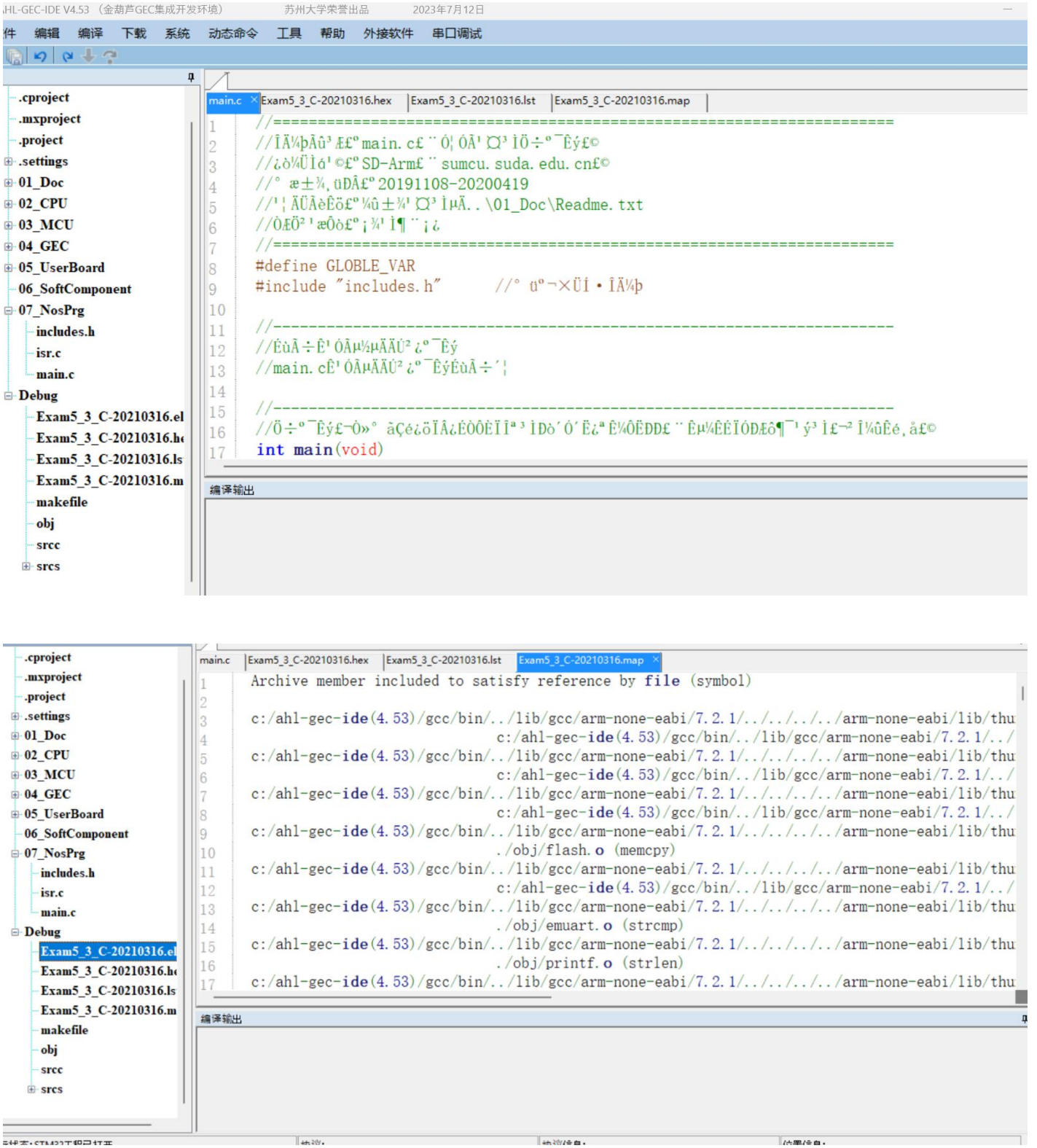


Figure 1. AHL-GEC-IDE project structure and main.c source file.

4. Source-Code Modification

The safest approach is to keep the original project framework unchanged and modify only the LED initialization and the LED state-control portion inside the main loop. The following version follows the same programming style as the supplied template.

4.1 LED initialization

```
/* Initialize both LEDs in the OFF state */
gpio_init(LIGHT_RED,   GPIO_OUTPUT, LIGHT_OFF);
gpio_init(LIGHT_BLUE, GPIO_OUTPUT, LIGHT_OFF);
```

4.2 Alternating BLUE and RED logic

```
/* Main loop: BLUE and RED alternate */
for(;;) {
    mMainLoopCount++;

    if (mMainLoopCount <= 12888999)
continue;

    mMainLoopCount = 0;

    if (mFlag == 'B')      {          /*
BLUE       ON,       RED       OFF       */
gpio_set(LIGHT_RED,           LIGHT_OFF);
gpio_set(LIGHT_BLUE, LIGHT_ON);

        mFlag  =  'R';       }       else
{              /*  RED  ON,  BLUE  OFF  */
gpio_set(LIGHT_BLUE,          LIGHT_OFF);
gpio_set(LIGHT_RED,  LIGHT_ON);

        mFlag =
'B';     }
}
```

Note: The value 12888999 is retained from the existing template style to create a visible delay without changing the overall program structure. If the supplied GPIO_02 template uses a different delay mechanism, keep that existing mechanism.

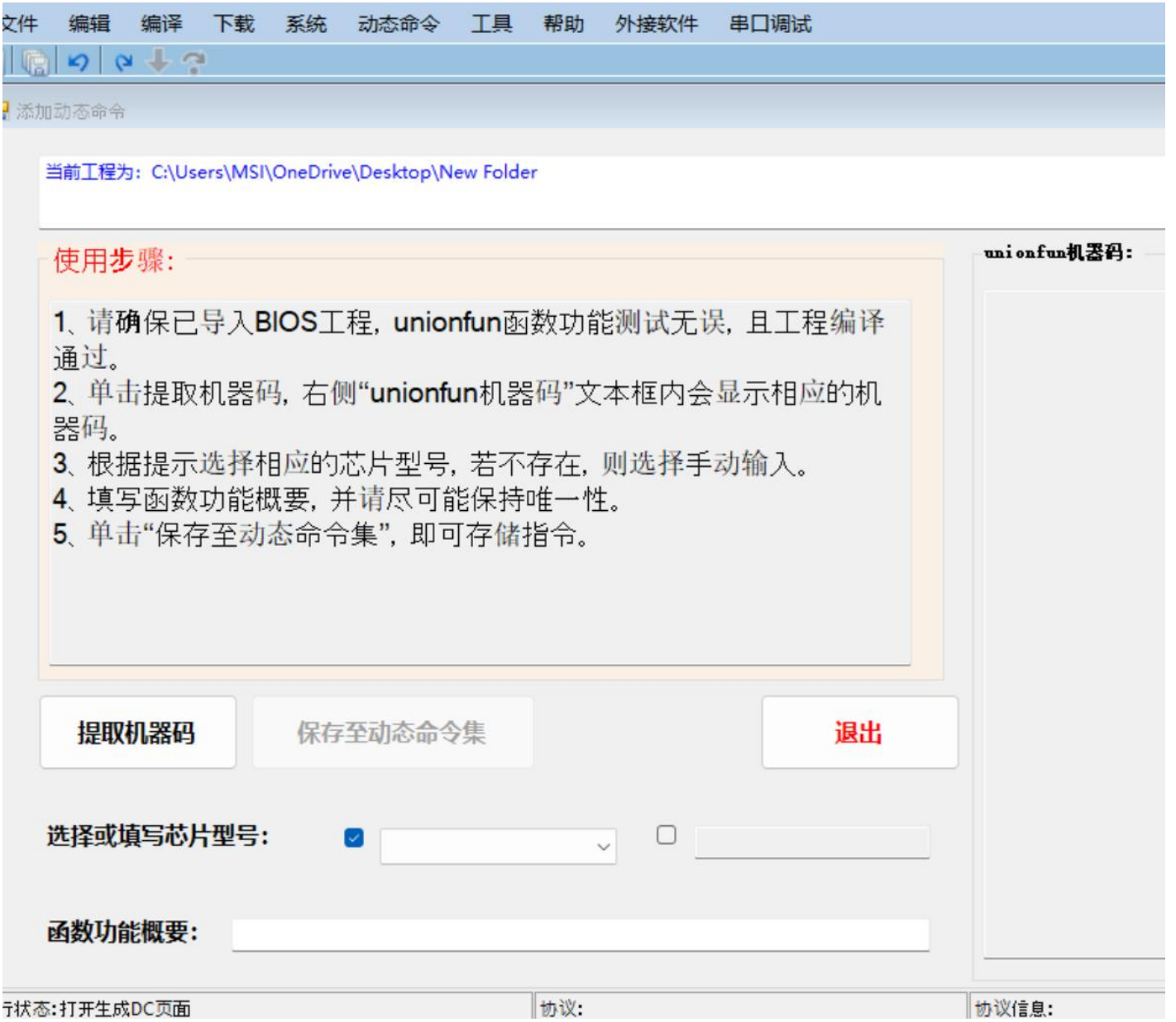
5. Experimental Procedure

1. Open the project — Open GPIO_02 in AHL-GEC-IDE V4.53.
2. Open source — Open 07_NosPrg → main.c.
3. Initialize LEDs — Set LIGHT_RED and LIGHT_BLUE as GPIO outputs and start them OFF.
4. Edit the loop — Replace the existing single-color LED switching logic with the alternatingBLUE/RED logic.
5. Save — Press Ctrl+S and save main.c.
6. Compile — Use → (Compile Project).
7. Check output — Confirm that the compile-output panel reports no errors.
8. Download — Use the IDE download/program function to transfer the program to theSTM32 board.
9. Run — Power/run the board and observe the LED sequence.
10. Verify — Confirm that BLUE and RED appear alternately and are not ON together.

6. Verification Logic

State	RED	BLUE	Expected output
B	OFF	ON	BLUE
R	ON	OFF	RED
B	OFF	ON	BLUE
R	ON	OFF	RED

7. IDE Evidence



BLUE LED	BLUE illuminates while RED is OFF.	Attach board photo after testing.
RED LED	RED illuminates while BLUE is OFF.	Attach board photo after testing.
Alternation	BLUE → RED → BLUE → RED continuously.	
Compilation	No compile errors.	

8. Results

Submission note: The report intentionally does not claim a physical LED result that has not been photographed. Add your actual RED/BLUE board photographs after the program runs.

9. Conclusion

The GPIO project was adapted to demonstrate alternating digital output control. The program switches between two mutually exclusive states: BLUE ON with RED OFF, then RED ON with BLUE OFF. This demonstrates GPIO initialization, output control, state flags, and repeated execution inside the main loop.